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ORIGINAL PAPER

A framework integrating affinity propagation algorithm and spatial bivariate analysis for enhanced identification and localisation of soil heavy metals pollution sources

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Abstract The accurate identification of pollutant sources and their spatial distribution is crucial for mitigating soil heavy metals (SHMs) pollution. However, the receptor model struggles to effectively categorize pollutant sources and pinpoint their locations and dispersion trends. We propose a novel comprehensive framework that combines a receptor model, random forest (RF), affinity propagation (AP) algorithm, and bivariate local indicator of spatial association (BLISA), to optimize the traditional approach for tracing SHMs sources in industrial regions. We apportioned SHMs sources using a receptor model combined with RF, while BLISA combined with AP methods were employed to accurately locate the source areas and identify their dispersion tendencies. The results revealed that SHMs originated from mixed sources of equipment manufacturing agglomeration and agricultural activities (59.0%), geological background (30.5%), and emissions from heavily-polluting industries (10.5%). The pollution sources of soil Cd and Pb were located near specific industries, showing characteristics of multi-site concurrent pollution diffusion influenced by their proximity to industrial sites. The spatial distribution of Cr, Cu, and Zn sources was concentrated in high-density urban industrial areas, transitioning from point to nonpoint sources, with diffusion patterns influenced by the spatial agglomeration effect of industries. Our enhanced framework accurately identifies the location of SHMs sources and their dispersion tendencies, thereby improving regional soil pollution management.

Keywords Soil contamination · Random forest · Spatial diffusion pattern · Industry agglomeration · Risk management

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Introduction

Soil safety is crucial for the human survival and development (Jia et al., 2020a, 2020b). In recent years, rapid urbanisation, industrialisation, and agriculture have led to soil contamination threatening the safety of soil environments (Facchinelli et al., 2001; Jia et al., 2020a, 2020b; Wang & Lu, 2011). Soil heavy metals (SHMs) have attracted much academic and public attention because of their biotoxicity, bioaccumulation, and degradation challenges, which have caused soil pollution and human health problems (Cloquet et al., 2006; Yi et al., 2023). Currently, 16.1% of the soil samples in China are polluted with SHMs. The Chinese government has proposed a national strategy for the risk management of SHMs pollution, with the goal of effectively controlling the risk of soil pollution by 2025.

The sources of SHMs are complex, and strengthening the prevention and control of pollution sources has become an effective way to reduce the environmental risks and cost of later treatment. Source apportionment of SHMs has been one of the most important topics in recent years (Wang et al., 2022). Several traditional receptor methods (e.g., absolute principal component score/multiple linear regression analysis [APCS/MLR], positive matrix factorisation [PMF], and chemical mass balance models [CMB]) have been widely applied to apportion source contributions, and are highly convenient and efficient (Anaman et al., 2022; Harrison et al., 1996; Hopke, 1985; Liao et al., 2021; Srivastava et al., 2005; Zhang et al., 2021a, 2021b; Zhou et al., 2023). However, the interpretation is based on an evaluator's judgement, which has the limiting characteristic of subjectivity. Moreover, the importance of the relevant influencing factors cannot be accurately identified, making it difficult to control local SHMs pollution and remediation (Lv and Liu, 2019; Shao et al., 2018; Zhang et al., 2021a, 2021b). To overcome this problem, many successful studies have been conducted using methods to objectively interpret and validate the results of receptor models by introducing environmental variables to identify the main factors influencing SHMs through machine learning methods (e.g., finite mixture distribution model, gradient boosting decision tree, and random forest [RF]) (Zhang et al., 2021a, 2021b). The RF model has been widely used because of its ability to quantitatively assess the importance of each environmental variable and its insensitivity to noise features, which can effectively avoid overfitting (Breiman, 2001; Shi et al., 2022). Yang et al. (2021) combine PMF and RF to identify the SHMs contamination sources and key environmental factors. Although RF has advantages in identifying the key factors and source analysis of SHMs, it is unable to identify the spatial clustering patterns and spatial interaction mechanisms between the influencing factors and SHMs concentrations. It also has difficulties in locating the pollution sources and grasping their potential spatial diffusion trends, which leads to serious challenges regarding the prevention and control of the SHMs pollution sources.

Currently, geostatistical analyses based on the first law of geography (e.g. Kriging interpolation) are commonly employed alongside source apportionment models to study the spatial distribution patterns of SHMs concentrations and pollution sources, to qualitatively assess the possible spatial relationship between the two, and provide approximate spatial information on pollution sources. However, spatial interpolation has an evident smoothing effect and tends to filter out unusually high SHMs concentrations and contributions that might be attributed to human activity (Liu et al., 2016). Moreover, spatial interpolation cannot quantitatively identify spatially clustered areas of SHMs contamination and pollution sources, making it difficult to accurately locate pollution sources. A bivariate local indicator of spatial association (BLISA) describes the spatial interactions between bivariate variables at the local scale, and serves as an important tool to characterise the spatial clustering patterns between SHMs concentrations and pollution-contributing factors (source–pathway–receptor) and identify the pollution sources. For example, BLISA can reveal the spatial clustering patterns of environmental factors (industrial activities) that have been screened using an RF model and SHMs concentrations, so as to identify priority areas of the SHMs pollution risks (Huang et al., 2022). However, it is difficult to locate the specific pollution sources and dispersion trends, which impacts the effective implementation of soil pollution control programmes. The graph theory-based affinity propagation (AP) algorithm, which can incorporate geographic distances into the spatial clustering process of pollution levels, can screen potential pollution source sites and portray the propagation and diffusion

trajectories of pollution in a space. AP has been successfully applied in the urban air pollution studies (Wang et al., 2020a, 2020b) but has not yet been utilised in the soil environment field. Moreover, there is a lack of integrated frameworks for a more systematic and accurate identification of contaminant source types, contribution rates, spatial clustering correlations between SHMs concentrations and contributing factors, potential pollution source locations, and their spatial diffusion patterns. Therefore, integrating the aforementioned frameworks can provide rich spatial information on pollution sources and further enhance the efficacy of source apportionment.

We select an industrial county in Jiangsu Province, China, as a case study. We use a novel framework to diagnose the contributing factors and locate the SHMs pollution sources in order to address the aforementioned issues and develop a theory of pollutant source apportionment (Fig. 1). In our framework, we combine APCS/MLR with RF to identify likely pollution sources and assess the contribution of environmental factors. We use BLISA and AP to locate the pollutant sources and trace their spatial diffusion patterns. Our objectives are to (1) apport potential SHMs sources using quantitative methods, (2) establish spatial clustering and autocorrelation patterns of SHMs concentrations and contributing factors, and (3) directly identify SHMs pollution sources and their diffusion patterns.

Materials and methods

Study area, soil sample collection, and chemical analysis

Jintan County is located in Changzhou City, Jiangsu Province, China (31°33′42″ N-31°53′22″

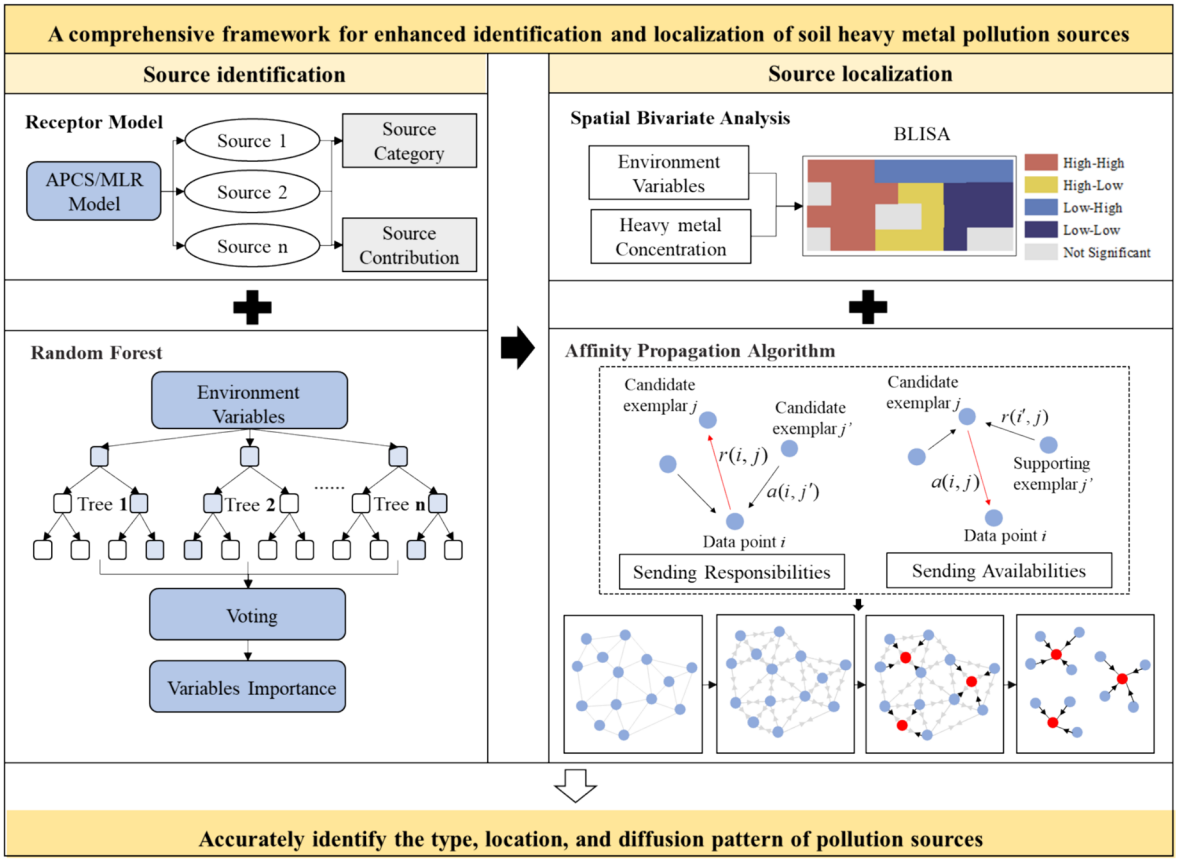


Fig. 1 The workflow of the proposed framework for source apportionment and localisation

N, 119°17′45″ E-119°44′59″ E) in the northwestern part of the Taihu Lake Basin (TLB), covering a total area of 975.68 km^2. It has a humid monsoon climate and lies in the northern subtropical zone, with abundant sunshine and rainfall. The average annual temperature reaches 15.3 °C, and the average annual precipitation amounts to 1063.5 mm. The topography of the territory gradually decreases from east to west, with the low hills of the Maoshan Mountains running north to south in the west and the alluvial lakeside plains of Taihu Lake in the east (Fig. 2a). The permanent landforms include low hills with bedrock, loess hills, as well as alluvial and lakeside plains. The Devonian sandstone distribution area located in the western Maoshan Mountains is one of the main Se-rich soil areas in Jiangsu. The area of agricultural land in Jintan County spans 619.12 km^2, which accounts for 63.45% of the total regional area. Rice soil is the main soil type, which accounts for 75.87% of the arable land area. Additionally, Jintan County has a well-developed economy and ranks the top 100 industrial counties in China. It has numerous factories covering industries such as automobiles and parts, new photovoltaic energy, textiles and garments, equipment manufacturing, chemicals, and new materials.

In this study, a total of 170 samples were collected from the top soil (0–20 cm) of Jintan County using a 2 km×2 km grid and a 5-point mixed sampling method in August 2020 (Fig. 2a). Soil samples were air-dried, ground and sieved to determine their soil physicochemical properties and SHM concentrations. In particular, soil pH was determined by the electrode method; soil organic matter was determined by the external heating method using potassium dichromate; and SHMs were determined by the tetra-acid digestion method. Weigh 0.1 g of soil samples and pass them through a 0.149 mm nylon sieve into a polytetrafluoroethylene beaker, add 37%HCl(3 mL)–65%HNO_3(1 mL)–65%HF(6 mL)–65%$HClO_4$(0.5 mL) mixed acid solution sequentially, seal the beaker and leave it overnight. Next, disintegrate the mixture under a high temperature of 180 ℃ for 10 h. After cooling, add 2 mL of aqua regia (mixed with 1 mL of water) to the evaporated transparent solid. Add 2 mL of aqua regia ($HCl:HNO_3$=3:1), and accurately titrate with deionised water to 10 mL for analysis in the above test. Cd, Pb, Cr, Cu, and Zn were determined by inductively coupled plasma mass spectrometry (ICP-MS; Elan 9000; PerkinElmer AB SCIEX, Redwood City, CA, USA). Hg, As and Se were digested with 5 mL aqua regia and 5 mL deionized water in a water bath at 100 °C for 4 h and

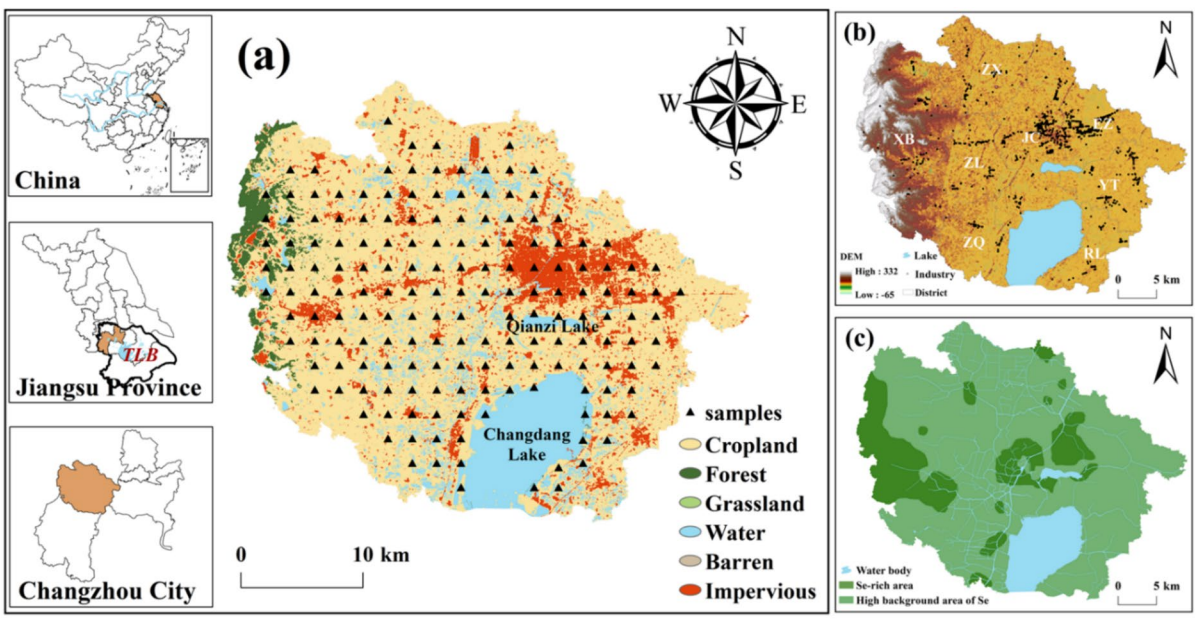


Fig. 2 **a** The map of geographical location, land-use type, and sampling sites; **b** digital elevation model, pollution industries; **c** soil Se concentration in the study area. Note: TLB, Taihu Lake Basin

then determined by atomic fluorescence spectrometry (AFS, Jitian AFS-820, Beijing, China). To ensure the accuracy of the detection, the standard reference materials (GBW07405, National Research Center for Certified Reference Materials of China), blank samples, and duplicate samples were tested after every five samples for correction. The recovery of all samples ranged from 85 to 115%.

Environmental data collection and processing

Considering the source, pathway, and receptor chain of contamination affecting SHMs concentrations, referred to as the "source–pathway–receptor" (S–P–R), a total of 12 covariates in three categories were selected as environmental covariates for this study (Table S1). Industrial points of interest were obtained from the Gaode Map Open Platform (https://lbs.amap.com/), road and river data were from OpenStreetMap (https://www.openstreetmap.org), and land use data were obtained from the 30 m x 30 m China Land Cover Dataset (CLCD) provided by the School of Remote Sensing, Wuhan University (Yang & Huang, 2021). A digital elevation model (DEM) was obtained from the Geospatial Data Cloud, and altitude and slope were calculated using DEM with spatial analysis. All environmental variables were processed in ArcGIS 10.8.

APCS-MLR model

The APCS-MLR model is widely used for quantitative SHMs source apportionment. It assumes that the total SHMs concentration is the sum of contributions from different sources (Harrison et al., 1996). In the APCS-MLR model, negative values may appear in the contribution calculation, which affects the accuracy of source allocation. Therefore, all negative values are taken as absolute values and converted into positive contributions to represent the SHMs contribution of the corresponding sources (Haji Gholizadeh et al., 2016). The specific calculation steps are as follows:

$$Z_{ik} = \frac{X_{ik} - \overline{X}_i}{\sigma_i} \tag{1}$$

$$P_{j0} = \frac{0 - \overline{X}_i}{\sigma_i} = -\frac{X_i}{\sigma_i} \tag{2}$$

$$(APCS)jk = Pjk - Pj0 \tag{3}$$

$$Xik = \sum_{j=1}^{n} C_{ij}(APCS)jk \tag{4}$$

where Z_{ik} is the standardised content of the element i in the sample k; X_{ik} represents the content of the element i in the sample k; $\overline{X}_i$ and σ_i are the mean content and standard deviation of element i, respectively; j denotes the number of factors; P_{j0} signifies the factor score of the samples with 0 content, which is utilized to compute the absolute factor scores$(APCS)_{jk}$; C_{ij} stands the linear regression coefficient of factor j on element i; The mean value of $C_{ij}(APCS)_{jk}$ over all samples is employed to represent the contribution of pollution sources.

Random forest

Random forest (RF) is a machine learning method proposed by Breiman (2001). In recent years, RF has been widely used in SHMs source analysis. The RF model is an extension of the decision tree aimed at improving its prediction performance. The error and variance of RF model are typically low, enabling it to avoid model overfitting. The RF algorithm was implemented using the RandomForestRegressor package in Python 3.8. In this study, the optimal parameters of the RF model are presented in Table S2. Because of the limited training samples, we used a Ten-fold cross-validation to verify the fitting accuracy of the model, while the model performance was evaluated using R^2 and RMSE, as shown in Table S3 and S4. More detailed algorithm principles of the RF model can be found in Breiman (2001).

Bivariate local *indicator* of spatial association (BLISA)

Spatial autocorrelation is used to detect spatial correlation in geographic data. BLISA is one of the local Moran metrics, which operates on the assumption that two closely related variables tend to show similar spatial patterns (Hu et al., 2021), was employed to analyze the spatial correlations between the SHMs concentrations and the contributing factors to identify agglomeration zones. The formula of the BLISA (I_{kl}) statistic can be defined as:

$$I\mathrm{kl} = \frac{X_k^j - \overline{XK}}{\sigma k} Wij \left(\frac{X_l^j - \overline{Xl}}{\sigma l} \right) \quad (5)$$

where X_k^j and X_l^j represent the values of k at i and j, respectively; $\overline{XK}$ and $\overline{Xl}$ denote the mean values of k and l, respectively; σk and σl indicate the variance of k and l, respectively; and W_{ij} represents the spatial weighting matrix indicating the proximity or distance between i and j. In this study, i and j represent the environmental factors and SHMs concentrations in 537 hexagonal grids with an area of 1 km^2, respectively. Furthermore, the pseudo-significance of the bivariate LISA statistic is evaluated at the 5% level based on 999 randomization permutations, which can effectively reduce the uncertainty. When $p \leq 0.01$, the relationship between factors and SHMs is extremely significant; when $p \leq 0.05$, it is considered significant, and when $p > 0.05$, it is not significant. The spatial correlations patterns were classified into five classes (high-high, high-low, low–high, low-low, and not significant) based on the results of BLISA. Additional details regarding BLISA can be found in Hu et al. (2021). BLISA was performed in Geoda v1.18.0.

AP algorithm

The AP algorithm, proposed by Frey and Dueck (2007), is a clustering algorithm based on data similarity. The fundamental concept involves considering all sample points as potential central cluster points, connecting pairwise lines between sample points to form a similarity matrix, and determining the central cluster points and clustering groups based on the attraction and belongingness of information transfer among network edges. The AP algorithm is noted for its lower squared error sum and superior clustering performance compared to other clustering algorithms.

First, we calculated the similarity between samples i and j is calculated to obtain the similarity matrix:

$$s(i,j) = -\|xi - xj\|^2 \quad (6)$$

where $s(i,j)$ represents the similarity between different sample points, with x_i and x_j represents the coordinates of the sample points i and j, respectively.

Second, we computed the attraction and attribution between sample points i and j:

$$r(i,j) = s(i,j) - \max_{j's.t.j' \neq j} \left\{ a(i,j') + s(i,j') \right\} \quad (7)$$

$$a(i,j) = \min \left\{ 0, r(j,j) + \sum_{i's.t.i' \notin \{i,j\}} \max \left\{ 0, r(i',j) \right\} \right\} \quad (8)$$

$$a(j,j) = \sum_{i's.t.i' \neq j} \max \left\{ 0, i', j \right\} \quad (9)$$

where $r(i,j)$ is responsibility, which measures the suitability of j as a central cluster point for i; $a\left(i,\dot{j}\right)$ is availability, which measures the suitability of i as a central cluster point for j; and $a(j,j)$ is self-availability. We iteratively updated the responsibility matrix, upon which the attribution matrix relied for information retrieval. To prevent data oscillation, a damping coefficient of 0.7 was set during the iteration process.

Drawing from previous studies (Li et al., 2023; Wang et al., 2020a, 2020b), we incorporate a similarity function that considered geographical and environmental variables:

$$D(i,j) = d_{dis}^{i,j} + d_{con}^{i,j} + d_{env}^{i,j} \quad (10)$$

where $D(i,j)$ is the generalised distance between i and j; $d_{dis}^{i,j}$ is the effect of the distance between sample points on the clustering results; $d_{con}^{i,j}$ is the concentration of SHMs; and $d_{env}^{i,j}$ denotes the environmental variables, which include Polluting industrial density, Distance of sample site from polluting industry, and so on. The AP algorithm was implemented using Python 3.10.7.

Results

SHMs concentrations and their spatial distributions

The descriptive statistics for the SHMs concentrations revealed that the average concentrations of Cd, Pb, Cr, Cu, Zn, Hg, As, and Se were 0.20, 32.51, 78.99, 26.64, 69.86, 0.11, 8.03, and 0.28 mg/kg, respectively (Fig. 3 and Table S5). The mean values, except for As, exceeded the background values of the Jiangsu Province soils to varying degrees. Compared to the Chinese soil quality standard (GB15618-2018), the exceedance rates of all SHMs were low, each being less than 1%. The Coefficient of variation (CV) of Cd,

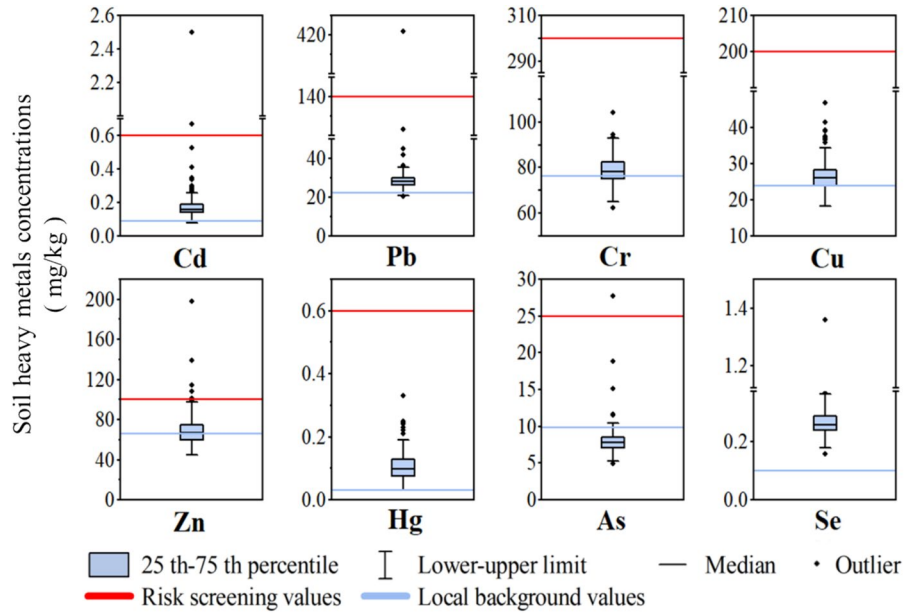


Fig. 3 Box plot of soil heavy metals concentrations (mg/kg)

Pb, Hg, and Se ranged from 37 to117% (Table S5), indicating a high degree of variability (> 36%), whereas Cr, Cu, Zn, and As exhibited relatively moderate or low variability (CV < 24%) (Lv et al., 2013). The single-factor pollution index (P_i) results indicated that the mean P_i values of Cd and Hg were 2.21 and 3.57, respectively, indicating a higher level of pollution, while the other elements were generally exhibited light or no pollution levels (Fig. S1). The P_i value of Se was elevated (2.8) due to naturally enriched Se soil in the area (Zhang et al., 2024).

There was evident spatial heterogeneity in the SHMs concentrations (Fig.S2). Cd, Pb, and Hg exhibited similar spatial distribution patterns, with hotspots exceeding the risk screening values predominantly situated in zones JC and XB, corresponding to urban construction areas. Conversely, the cold-spot zones were observed in the peripheral farmlands in the northern, southern, and western parts of the region. Cr, Cu, and Zn had similar spatial distribution patterns, with the hotspots of super-risk screening values concentrated in zones JC and EC, characterized by industrial and agricultural agglomerations with dense road networks, numerous factories, and extensive construction and agricultural land. The areas with elevated levels of As and Se were predominantly located on the periphery of Qianzi Lake in the central region and in the Se-rich area of the Maoshan Mountains in the west.

Source apportionment of SHMs

Source apportionment by APCS/MLR

The APCS/MLR model was employed to identify the sources of SHMs and quantify the contribution of each source. Prior to conducting the principal component analysis (PCA), we performed the Kaiser–Meyer–Olkin (KMO) and Bartlett's tests. The results yielded values of 0.597 and 389.59 (degree of freedom [df] = 28, $P < 0.001$), respectively. According to the criterion of eigenvalue > 1, three principal components were extracted, accounting for 67.8% of the total variance (Table S6). The R^2 of the APCS/MLR model ranged from 0.51 to 0.83, indicating a high level of model fitting accuracy. The APCS/MLR model identified three source factors: F1, F2, and F3, which accounted for 59.0%, 30.5%, and 10.5% of the total variance, respectively. Among them, Cr, Cu, and Zn were the most affected by F1, with contributions of more than 93%. This was followed by Hg and Pb, with contributions of 69.0% and 63.1%, respectively. F2 contributed the most to As (79.9%) and Se (72.3%), and made a significant contribution to Cd (50.0%). Additionally, F3 exerted a significant influence on the pollution levels of Hg and Cd, contributing 29.1% and 30.2%, respectively (Fig. 4 and Fig. S3).

Key factors affecting pollution sources in RF

To reduce the subjectivity and uncertainty with identifying the sources of SHMs pollution, an RF model was employed to evaluate the impact of each environmental variable on the SHMs pollution sources. Based on the evaluation results of the prediction accuracy by RF model, the root mean square error (RMSE) of F1, F2, and F3 were 0.29, 0.37, and 0.35, respectively, and the R^2 between the predicted and observed output values were 0.91, 0.86, and 0.88, respectively (Table S3). These results suggest that the constructed RF model performed effectively. Figure 5 illustrates the relative importance of each environmental variable for each source factor in the RF model. Greater importance values indicate stronger influence of the environmental variables on the source factors. For F1, soil organic matter (SOM), and pollution industrial density (PID) exhibited the highest importance among the environmental variables, and the partial dependence plot revealed a positive correlation between SOM and PID with the F1 score. F2 was primarily influenced by SOM and pH, with the score increasing with SOM and decreasing with pH. For F3, SOM and the distance from polluting industries (Dis_ind) were identified as the two most significant environmental variables, with F3 showing a positive correlation with SOM and a negative correlation with Dis_ind (Fig. 5).

The spatial correlations between key influencing factors and the SHMs concentrations

Fig. S5 illustrates the influence of each driving factor on the concentrations of SHMs. Polluting industries, SOM, pH, and topography had significant effects on the concentrations of the SHMs. SOM and Dis_id were the primary factors influencing the concentrations of Cd, Pb, and Hg, contributing approximately 15–20%. The key environmental factors influencing Cr, Cu, and Zn were SOM content and PID. Soil physicochemical, topography, and rivers, representing natural factors, exerted the greatest influence on As

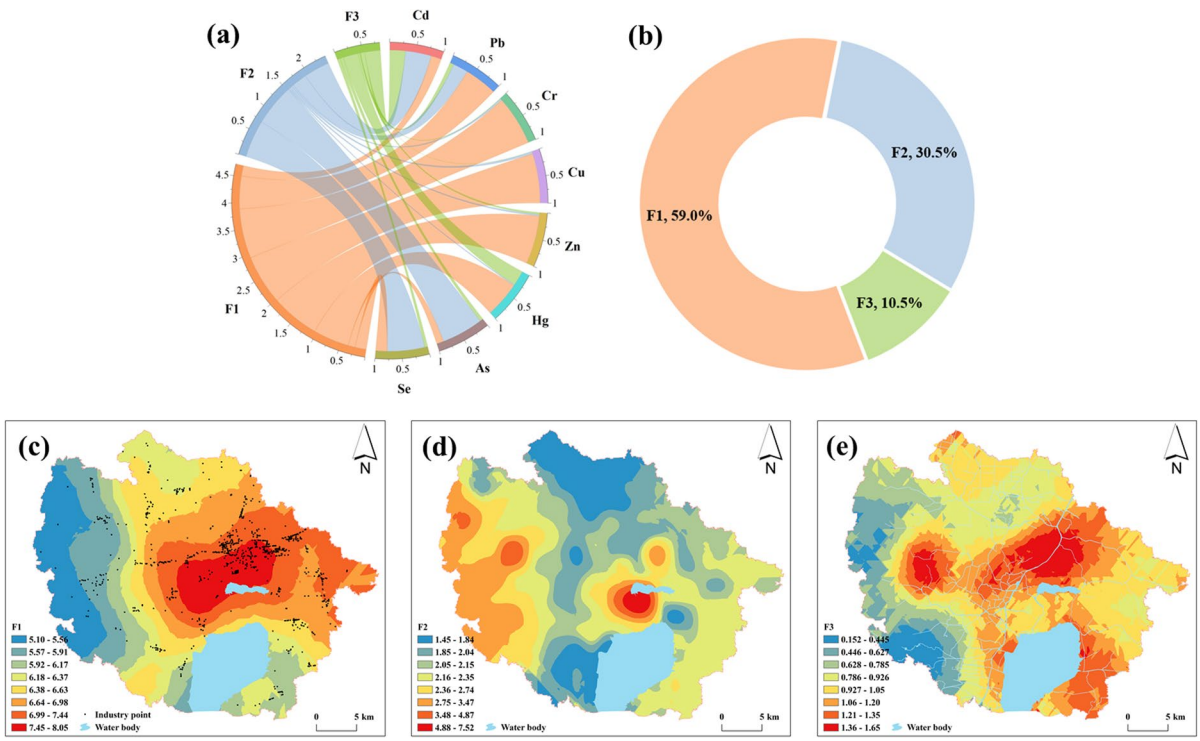


Fig. 4 Source apportionment of soil heavy metals based on APCS/MLR model. **a** Contributions of different sources to soil heavy metals; **b** contribution percentage of each factor; **c**–**e** spatial distribution of each factor

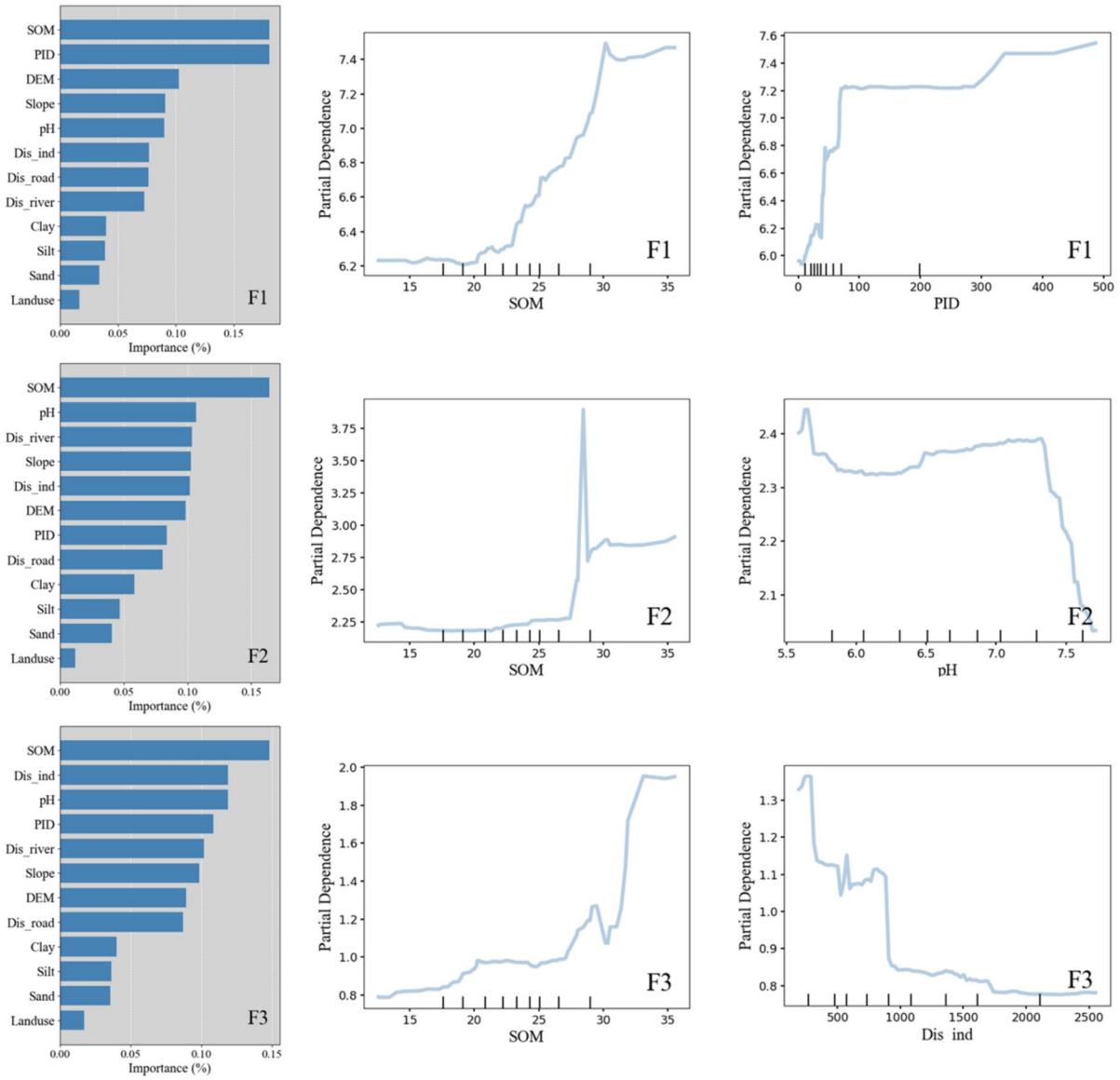


Fig. 5 The importance of the environmental variables on three source factors by RF model. Note: SOM, soil organic matter; PID, pollution industries density; DEM, Altitude; pH, soil pH; Slope, Slope; Dis_ind, Distance of sample site from polluting industry; Dis_road, Distance of the sample point from the road; Dis_river, Distance of the sample site from the river; Clay, Soil clay content; Sand, Soil sand content; Silt, Soil silt content; Landuse, Landuse type

and Se, contributing to a cumulative total of 53% and 57%, respectively.

Based on previous findings, we employed BLISA to analyse the spatial correlation patterns between the first two key factors and the SHMs concentration in order to delineate areas of SHMs contamination driven by different pollutant sources (Fig. 6). All the results passed the pseudo-significance ($p < 0.05$) of the BLISA statistic (Fig. S7 and S8). The high-high and low–high regions of Dis_id with Cd, as well as Dis_id with Pb, show greater consistency and are primarily situated in the central part of the study area. Additionally, a significant pollution hotspot was evident in a small region in the western part. In the low–high zone, the average concentrations of Cd and Pb were 0.21 and 38.3 mg/kg, respectively,

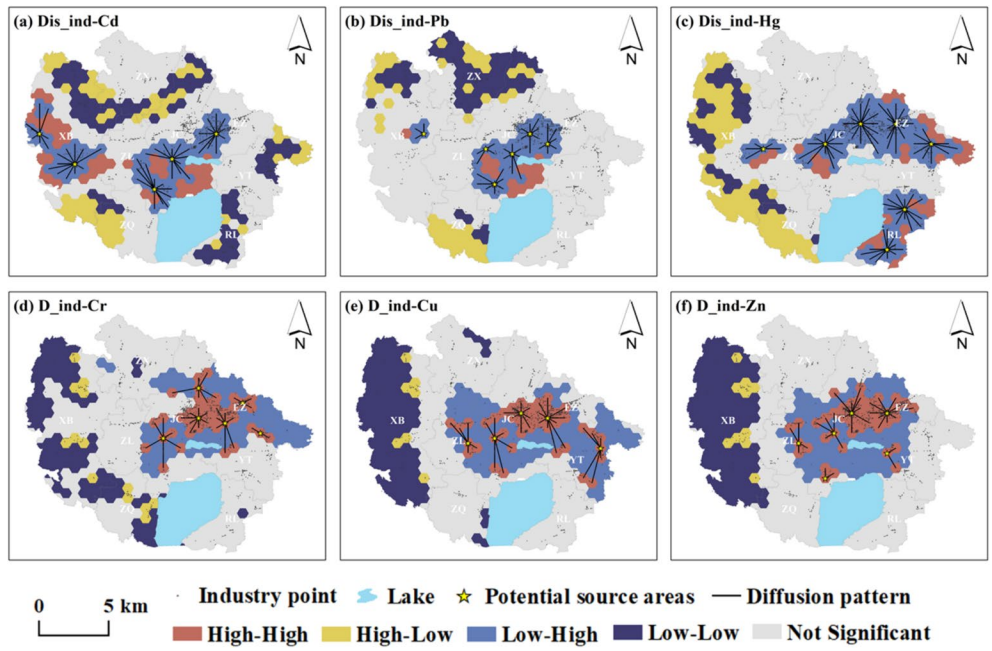


Fig. 6 Bivariate local indicator of spatial association (BLISA) map of environmental factors and soil heavy metals; potential source areas and diffusion patterns. Note: Dis_ind, Distance of pollution industry; D_ind, Pollution industry Density

exceeding the background values of soil in Jiangsu Province by 2.33 and 1.72 times. These regions were not industrial agglomerations within Jintan County but containing scattered polluting industries, including small- and medium-sized chemical factories and metal smelting plants. These industries dispersed pollutants over long distances through the atmosphere and waterways, experiencing minimal impact from the PID. The high-high and low–high zones depicting the Dis_id with Hg were predominantly concentrated in the central urban and southeast regions, with concentrations 4 times higher than those observed in Jiangsu.

The high-high zones between PID and the concentrations of Cr, Cu, and Zn demonstrated similarities and were predominantly located in the central manufacturing cluster. The manufacturing scale effect tended to exacerbate SHMs pollution levels. The low–high zone was mainly situated in the periphery of the high-high zone, influenced by spatial neighborhood and spillover effects. This area also constituted a zone with high SHMs pollution levels, comprising the largest proportion. These areas encompassed significant farmland, where agricultural activities could result in elevated concentrations of Cr, Cu, and Zn. SOM exhibited the highest percentage of high-high zones (Fig. S6), with concentrations of Cr, Cu, and Zn primarily distributed in the central and eastern farmland areas.

Identification of potential pollution source areas and spatial diffusion patterns via the AP algorithm

The concentrations of As and Se in the soil of Jintan County were primarily determined by the regional geological background, with no evidence of contamination. Therefore, except for As and Se, we identified the source areas for Cd, Pb, Hg, Cr, Cu, and Zn in the soil, which were contaminated by industrial activities playing a significant role. We used BLISA to explore the relationship between industrial activities and SHMs, and all the results passed the pseudo-significance ($p < 0.05$) of the BLISA statistic, and the significance is shown in Fig. S7 Based on the pollution industrial density delineated by BLISA, the high–high zones of the SHMs, as well as the distance between industries and the low–high zones of the SHMs, we used the AP algorithm to cluster their cores as potential pollution sources. The radiating directions towards neighbouring units within the

settlements were considered as the potential diffusion directions of the SHMs (Fig. 6). Areas with potential soil pollution sources of Cr, Cu, and Zn were relatively concentrated in industrial agglomeration zones. There were 5–6 pollution source areas with a diffusion range of 1.5–4.5 km. Owing to the influence of surface runoff and terrain, their diffusion paths exhibited a north–south trend. In comparison, the potential pollution source areas of Cd, Pb, and Hg were more dispersed, with sporadic occurrences in the western, southeastern, and central parts. These areas were predominantly occupied by heavily-polluting industries, such as the chemical industry, metal smelting, cement manufacturing, coal-fired power plants, and the textile industry. The distances between the pollution source areas were relatively large, with larger diffusion ranges (> 5 km), indicating a more complex diffusion pattern influenced by the interactions between the neighbouring potential pollution source areas. Particularly, Hg exhibited simultaneous pollution diffusion at multiple locations, indicating a greater diffusion risk. Furthermore, the northern and eastern parts of the largest lakes in the study area were threatened by the diffusion of Cd, Pb, and Hg. These SHMs can easily enter lakes through rainfall runoff or groundwater leaching, posing a threat to the water environment safety of lakes, which should be given sufficient attention. Overall, we found an overlap in the potential pollution source areas of these metals in the central urban area, suggesting the combined pollution from multiple metals had occurred in the region.

Discussion

SHMs source interpretation

F1 accounted for 59.0% of the total SHMs contribution (Fig. 4a and b), with a significant contribution of Cr (95.4%), Cu (95.3%), and Zn (93.3%) and a notable influence of Hg (69.0%) and Pb (63.1%) (Fig. S3). High factor scores were mainly distributed in low-altitude areas in the central and eastern parts, whereas lower scores were observed in the western mountainous regions (Fig. 4c). The correlation analysis indicated significant correlations between Pb–Cu, Cr–Cu, Cr–Zn, and Cu–Zn (Fig. S4), which suggested common pollutant sources. Over 75% of the industries in Jintan County are concentrated in the urban areas of the central and eastern regions. Studies have shown that the soil concentrations of Cr, Cu, and Zn in urban areas are strongly influenced by industrial activities, such as battery and electronic component manufacturing. With rapid industrial development in recent years, the automotive and equipment manufacturing industries in Jintan County have achieved considerable output values, reaching RMB 28.737 billion and RMB 29.778 billion, respectively, by 2020 (Statistical Bureau of Jintan County, 2021). These industries' production processes generate solid waste, sludge, and wastewater, leading to Cr, Cu, Zn, and Hg concentrations and SHMs pollution (Ali et al., 2016; Choppala et al., 2013; Hu et al., 2018; Wang et al., 2020a, 2020b). We found that the density of polluting industries was a major factor affecting F1, indicating the significant influence of industrial agglomeration on F1 (Fig. 5). Our observations revealed that the SHMs concentrations in the industrial agglomeration areas of Jintan County were significantly higher than those in other areas (Fig. 4c) due to increased energy consumption resulting from manufacturing agglomeration (Virkanen, 1998), leading to more environmental pollution. This finding is consistent with the findings of Yang et al. (2021). Thus, F1 is associated with the equipment manufacturing agglomeration. However, Jintan County is a traditional agricultural area with large areas of farmland in the central, northern, and eastern parts. In 2020, approximately 49.08 tons of pesticide packaging waste and 30.66 tons of plastic film were collected from the County (Statistical Bureau of Jintan County, 2021). Pesticides, fertilisers, and plastic film can increase the soil concentrations of Cr, Cu, Zn, and other SHMs (Liu et al., 2023), which is consistent with the findings of Qu et al. (2020). RF results also revealed that SOM, which reflected agricultural activities, were important factors that impacting F1. The previous study has reported that the central and eastern parts of Jintan County are hotspots for nitrogen and phosphate fertilisers input, resulting in high SOM, and SHMs concentrations alongside soil (Zhang et al., 2020). Moreover, SOM has numerous adsorption sites capable of chelating complex trace elements, further contributing to the accumulation of SHMs. Therefore, F1 is also associated with equipment manufacturing agglomeration and agricultural production activities.

F2 accounted for 30.5% of the total SHMs contribution (Fig. 4a and b) and mainly influenced As

(79.9%), Se (72.3%), and Cd (50.0%). As and Se exhibited significant correlations, whereas Cd showed significantly correlated with As and Se (Fig. S4). The average As concentration in the soil in Jintan County was lower than that of the soil background value in Jiangsu Province (Table S5), indicating minimal influence from human activities. Previous study in Jintan county shown that As was influenced by geological background (Qu et al., 2020). Jintan County is an Se-rich area, with the distribution of Se-rich soils closely related to the geological background of the Devonian sandstone distribution area in the western Maoshan Mountains (Fig. 2c). Zhang et al. (2024) have also found that Se-rich soils in the TLB are mainly controlled by geological background, climatic factors, and soil physicochemical properties, with weak human activity influence. Additionally, we found that the high-value areas of F2 were highly consistent with the distribution of Se-rich soils in Jintan County and were dominated by natural factors such as SOM, pH, and rivers (Fig. 5). The phenomenon of increasing F2 scores with increasing altitude also indicates the influence of geological background, as Se-rich soils are mainly located in western high-altitude mountainous areas with minimal human disturbance. We also found that some areas around Qianzi Lake (in the central area) had high scores, and overlaying these areas with the distribution map of the Se-rich soils revealed that these areas were all high Se background areas (Fig. 1c). F2 also contributed to 50% of the Cd in the soil. Although high variability of Cd are mainly caused by external anthropogenic activities, previous studies have also found a tendency for the relatively synchronous enrichment of Cd in Se-rich soils in Jiangsu Province. There is a highly significant positive correlation ($p < 0.01$) between Cd and Se, with high concentration values distributed in acidic Se-rich soils in the central and western regions, where Cd concentrations can reach up to 0.62 mg/kg. This reflects the natural geochemical affinity between Se and Cd (Liao et al., 2020). Therefore, F2 is geological background.

We found that F3 accounted for 10.5% of the total SHMs contribution (Fig. 4a and b) and primarily influenced Hg (29.1%) and Cd (30.2%) (Fig. S3). Hg and Cd concentrations reached moderate pollution levels and showed high variability, indicating anthropogenic point-source pollution. Jintan County is well developed industrially and has a large demand for thermal electricity. Some small, medium, and large coal-fired power plants are distributed within the area, with coal consumption exceeding 500,000 tons (Qu et al., 2020). The average concentration of Hg in coal is 0.17 mg/kg (Zhang et al., 2012), ranking second in its environmental contribution to Hg emissions during combustion. Among the many coal-fired power plants, a significant number have capacities below 200 MW, are equipped with low-efficiency subcritical boilers, and lack pollution control devices, resulting in Hg emission intensities over 10 times those of ordinary coal-fired units (Gao et al., 2019). Because of its high volatility, most Hg is emitted into the atmosphere during combustion, transported over long distances, and enters the soil through dry and wet deposition. Hg emission reduction flux ranked first in Jiangsu Province after phasing out high-energy-consuming coal-fired power plants during the 12th Five-Year Plan period (2011–2015) (Zhou et al., 2019). Therefore, it can be inferred that F3 is related to coal combustion emissions. It is worth emphasising that, unlike F1, which was influenced by pollution industrial density, the primary control factor for F3 was Dis_ind (Fig. 5), indicating that F3 is mainly affected by specific types of high-pollution industries (such as coal-fired power plants) rather than the agglomeration effect of equipment manufacturing. Overlaying the polluting industries with the F3 scores revealed that many chemical, coal-fired power plants, textile, papermaking, and metal smelting industries were distributed in the high-scoring areas of F3. These industries discharged large amounts of Hg and Cd into the surrounding environment during production activities (Lv and Liu, 2019; Zhou et al., 2019). The high-value F3 areas were mainly distributed in areas with dense river networks. This is because the aforementioned polluting industries generate large amounts of solid and liquid wastes containing Cd and Hg during production, which are discharged into rivers and subsequently entered the soil through irrigation and rainfall. Therefore, F3 represents emissions from heavily-polluting industries (i.e., coal combustion and chemical and smelting industries).

Spatial diffusion patterns of the SHMs pollution sources

The spatial diffusion patterns of the regional SHMs pollution are influenced by a combination of

pollution sources, diffusion pathways, and receptor attributes (Qiao et al., 2023). In terms of potential pollution sources, we found that industrial emissions were the primary SHMs pollution sources in the study area. However, different industries exhibited significant variations in the intensity and types of pollutants emitted. For soil Cd, Pb, and Hg in the study area, the distance from the polluting industry was the primary controlling factor influencing their input sources and showed the highest pollution level (Fig. S5). However, the potential pollution source areas did not overlap with regional industrial agglomeration areas but were scattered in areas where several heavily-polluting industries were distributed, such as coal-fired power plants, chemical plants, and metal manufacturing facilities (Fig. 6), which often had a greater impact on soil Cd, Pb, and Hg pollution than other industry types (Jia et al., 2019). The traditional view suggests that industrial agglomeration leads to increased productivity and energy consumption, which causes regional environmental pollution (Virkanen, 1998). However, some studies have suggested that the soils surrounding dispersed small and medium-sized chemical and metal smelting industries often develop into pollution hotspots, as these industries frequently lack effective supervision and pollution control measures (Yang et al., 2021). However, we found that the impacts of these factors on soil pollution diffusion were not fixed. For example, in the northern part of the study area, there were low–low areas for Cd and Pb. Although these were close to polluting industries, they did not constitute pollution sources. This may have been influenced by other factors, such as receptor attributes, as these areas were spatial interaction zones with low SOM, Cd, and Pb (Fig. S6 and S8). Studies have shown that it is difficult for soil with low SOM content to effectively adsorb SHMs and fix them in the soil, resulting in their diffusion in soluble forms (Kou et al., 2024). The central part of Jintan County is surrounded by high terrain on three sides and has a developed river network known as a flood corridor. Because Cd, Pb, and Hg were more active in the study area, they could spread over long distances through flood events, long-distance surface runoff, as well as atmospheric deposition. Therefore, compared to the other elements, the diffusion ranges of Cd, Pb, and Hg in the study area were larger, threatening the safety of regional lake water environments. This requires urgent attention from relevant regulatory authorities.

The formation of the diffusion patterns of Cr, Cu, and Zn was predominantly influenced by the industrial agglomeration effect (Fig. 6), aligning with the findings of Virkanen (1998) on the causes of environmental pollution. The AP algorithm identified the central part of Jintan County as a potential pollution source for Cr, Cu, and Zn. This high-tech industry agglomeration area includes automotive parts, equipment manufacturing, and photovoltaic industries (Fig. 6). This finding was consistent with the high–high areas between SOM and the above SHMs concentrations, indicating that higher SOM levels increased the likelihood of an area being a pollution source (Fig. S6). Owing to the scale effect of industrial agglomeration and the urbanization of their locations, a strong correlation between pollution sources, resulting in the synergistic spread of pollution and nonpoint source pollution trends in the surrounding soil. For example, the diffusion paths of Cr, Cu, and Zn connected the high–high and low–high areas. In terms of the diffusion direction, these three elements primarily migrated from the north and northwest to the south and southeast. Considering the distribution characteristics of the terrain and rivers in Jintan County, the rivers generally flow from north–south, making it easier for Cr, Cu, and Zn to migrate and diffuse towards the south. Moreover, we noticed small areas of high–low distribution in the west. Despite the high density of polluting industries, the concentrations of Cr, Cu, and Zn were low. This phenomenon is attributed to the transition zone from the western mountainous region, characterized by steep slopes, rapid surface runoff, and susceptibility to soil erosion. Consequently, pollutants generated by industrial activities migrate to low-lying and less sloping areas, reducing the contribution of soil pollution from these industrial agglomeration areas.

Limitations and implications

In previous studies, the identification of SHMs pollution sources has been accomplished through the use of receptor models and machine learning (Jia et al., 2020a, 2020b; Yang et al., 2021). However, they have failed to provide specific localisation information or further subdivisions of the source types.

These limiting factors and the lack of precise spatial information have become challenges in formulating soil pollution source control plans and policies. To address this issue, some scholars have used the PCA-Geodetector-MLR method to explore the SHMs pollution sources (Zeng et al., 2022). Additionally, Li et al. (2012) revealed the isolated aggregated spatial pattern of soil pollution and validated the feasibility of network and spatial cluster analysis in predicting potential pollution diffusion trends based on the graph theory. However, previous studies have only considered the concentration of SHMs and have neglected to acknowledge the importance of environmental factors in the diffusion of SHMs. In this study, we first applied the APCS/MLR model to analyze the sources and contributions of SHMs pollution, and used RF to enhance the accuracy of source identification. Furthermore, we incorporated BLISA and AP into the source analysis framework. Based on the spatial clustering patterns of different pollutants and SHMs, we combine the similarities of different environmental and geographic variables to obtain clustering results with a small squared error, thereby simulating and characterising the diffusion patterns of the SHMs pollution in complex urban and suburban environments. Our study can compensate for the uncertainty of the traditional source analysis methods by accurately and intuitively identifying the SHMs pollution hotspots, potential pollution sources locations, and the diffusion patterns. This is helpful for subsequent risk assessment and implementation of risk control strategies (Rinklebe et al., 2019).

However, our framework still has several limitations. Firstly, we were unable to conduct quantitative analysis of agricultural nonpoint source pollution. Due to limited data availability, we did not include indicators related to agricultural sources. Instead, we employed 12 environmental variables to evaluate the factors influencing SHMs pollution. However, this approach might not accurately discern the specific contributions of agricultural activities to SHMs and pollution sources. In future work, we plan to explore methods for selecting agricultural grid data suitable for county-scale analysis and augmenting the number of environmental variables to enhance the accuracy and applicability of our method. Secondly, our analysis predominantly relied on qualitative methods to discuss the factors influencing the diffusion of SHMs. We have not yet comprehensively and quantitatively revealed the formation mechanism of pollution source diffusion trends from the perspective of the "source-pathway-receptor" (S-P-R) theory, nor quantified the intensity of their effects. Hence, future work should utilize advanced geographic spatial analysis tools for quantitative analysis, including spatial econometric models (SEM), multiscale geographically weighted regression (MGWR), and geographic detector models (GDM). Thirdly, due to the significant spatial heterogeneity of regional soil environments and human activities, the applicability of our framework to other regions and spatial scales requires further validation, and the underlying theory also necessitates refinement. Nonetheless, our findings demonstrate the efficacy of combining AP algorithm with BLISA methods for tracing SHMs sources.

Conclusion

We combined the APCS/MLR, RF, BLISA, and AP methods to apportion and locate the SHMs pollution sources in Jintan County, an industrially developed county in Jiangsu Province, China. The results of the combined APCS/MLR and RF methods revealed that the SHMs primarily originated from mixed sources of equipment manufacturing agglomeration and agricultural activities (59.0%), geological background (30.5%), and emissions from heavily-polluting industries (10.5%) (i.e., coal-fired power plants, chemical plants, and metal smelting). We then integrated the BLISA and AP methods to identify the SHMs pollution sources. The results indicated that the diffusion patterns of soil Cd, Pb, and Hg were primarily controlled by the proximity of key pollution industries. These pollution sources were dispersed in the central, western, and southeastern industrial areas, indicating the characteristics of concurrent pollution diffusion in multiple areas and revealing a relatively complex diffusion pattern. Influenced by surface runoff and atmospheric deposition, their diffusion ranges were extensive, posing a threat to the environmental safety of the lakes in the southern region. The diffusion patterns of soil Cr, Cu, and Zn were mainly influenced by the spatial agglomeration effect of industries, with potential diffusion source areas concentrated in the high-density central urban industrial areas. However, the diffusion range was relatively small, and there was a gradual transition from point to diffuse

sources. Influenced by the terrain, the diffusion direction was from north–south. Overall, our framework provides new insights for enhancing the apportionment and management of SHMs pollution sources on a regional scale. The future studies should implement this framework in larger geographic areas and diverse regions.

Author contributions Zhang Feng contributed to investigation, laboratory, methodology, and writing—original draft. Shenglu Zhou contributed to conceptualization. Zhenyi Jia contributed investigation, writing—review & editing draft, funding acquisition, and supervision. Xuefeng Xie contributed to methodology, and laboratory analysis. Mingxing Xu contributed to data curation, visualization, and funding acquisition. Shaohua Wu contributed to writing—review & editing draft, supervision. All authors reviewed the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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